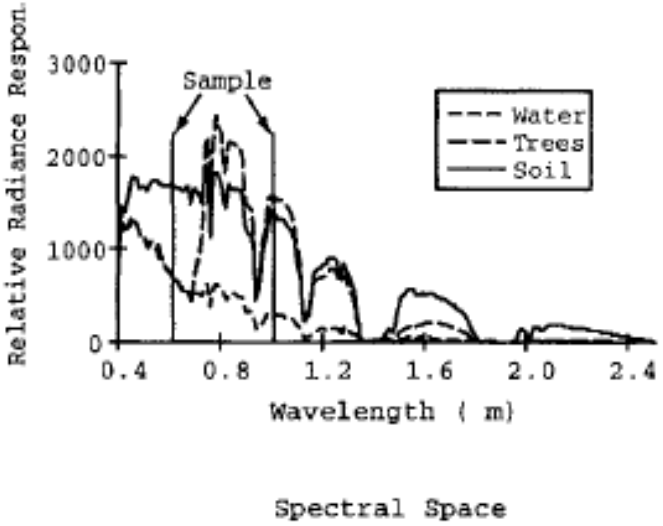
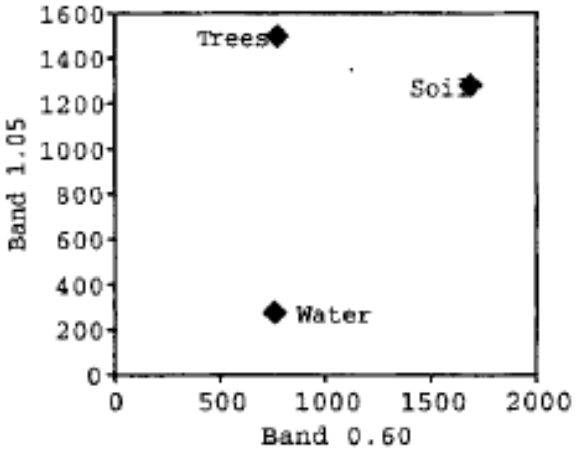
Image, Spectral and Feature Space

RS data in Image, Spectral and Feature Space





Spectral Space



Feature Space

Image Space

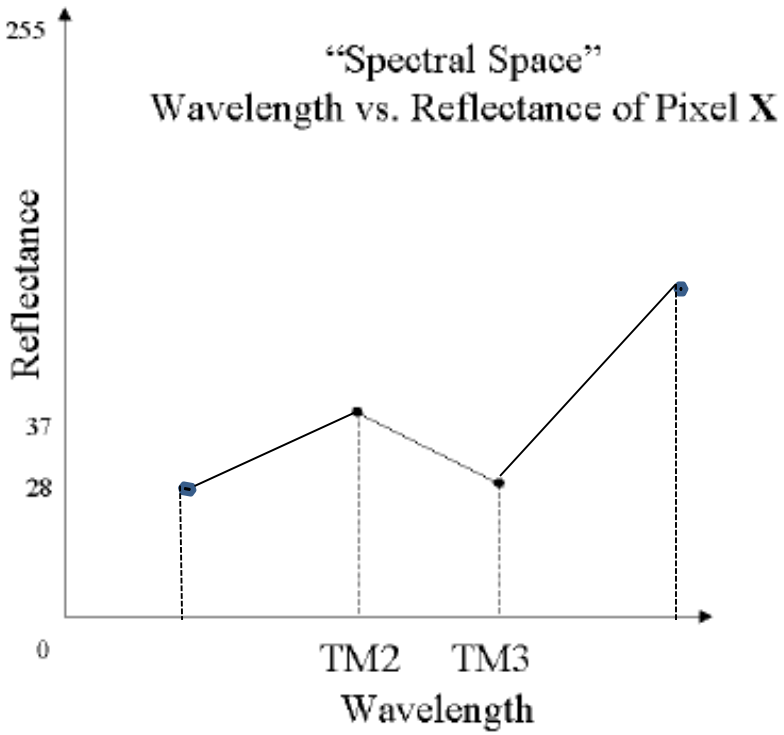
(Spatial + Spectral)

Feature Space

- **Spectral Space description allows us to describe one pixel only at a time**
- **A “Feature Space” plot satisfies this need where reflectance in one band is plotted against reflectance in another band**

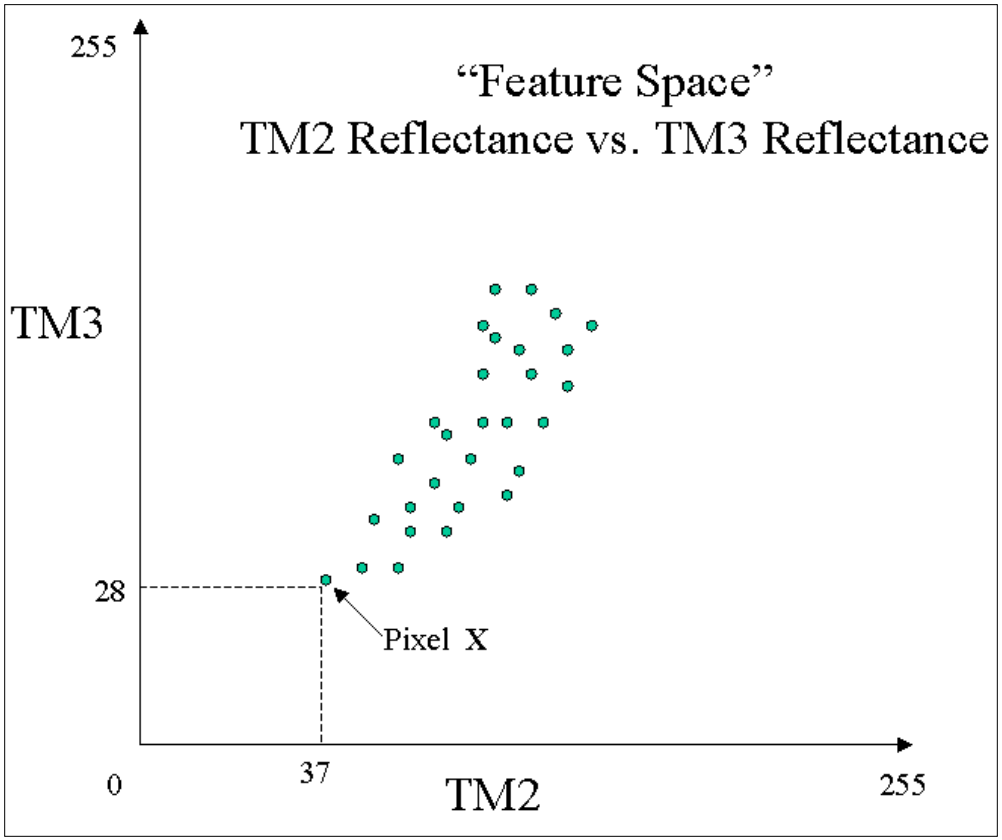
Spectral Space

- Consider a pixel X, in a Landsat TM image
- This pixel may be described by its spectrum using its reflectance in the wavelengths range of Bands 2 and 3
- Describing an object using multiple wavelengths of light, such a spectrum, is said to be a spectral space

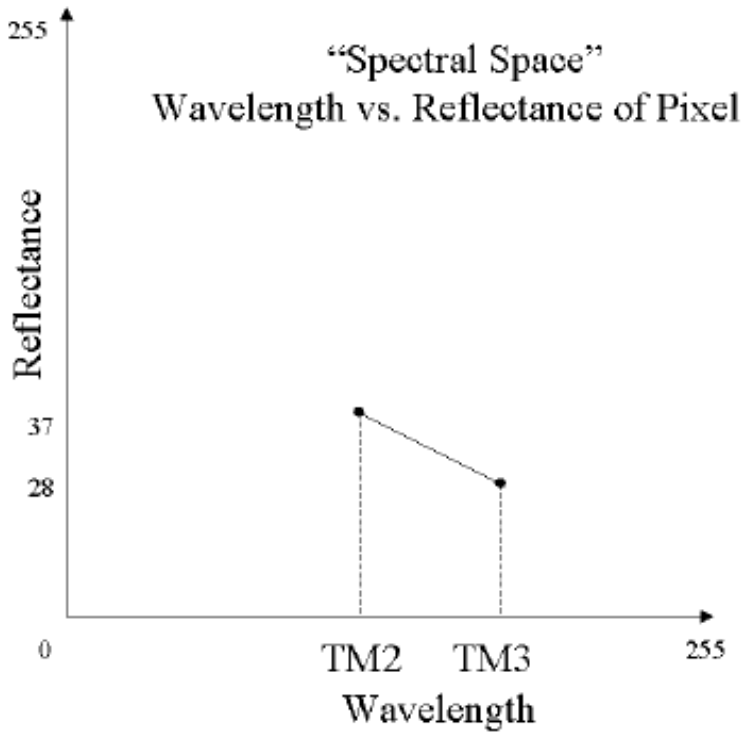


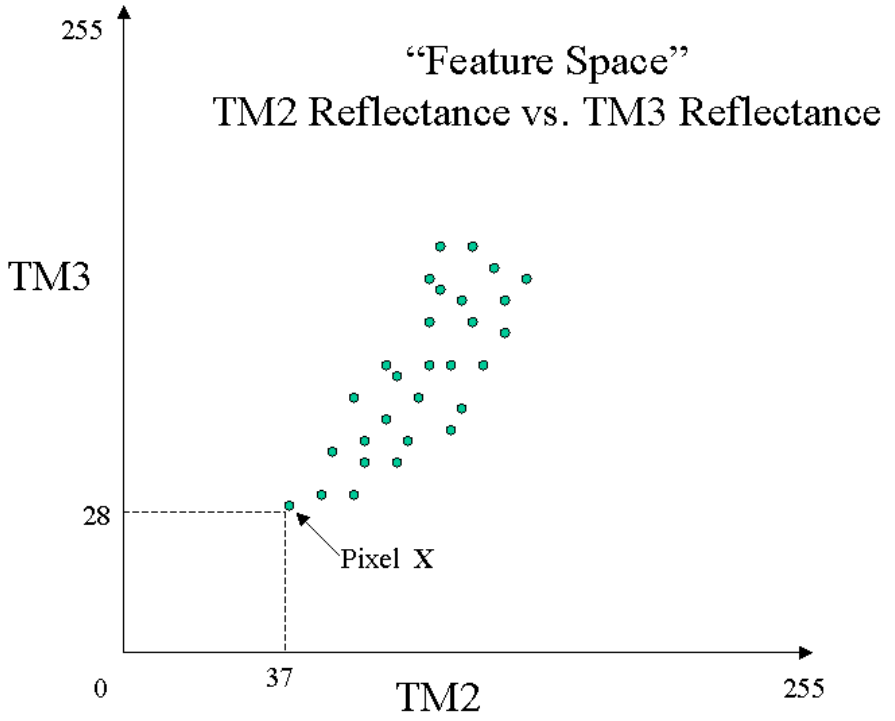
Feature Space

- **Feature Space,** a pixel X plotted along side several other pixels
- Data in feature space helps to
  - Visualize spectral relationships in data
  - See what is spectrally similar or dissimilar
- In this case, we can see that
  - Variation in Band 2 is related to variation in Band 3
  - Data in the two bands are correlated



Spectral Vs Feature Space





Spectral Redundancy

- A tendency for multiband datasets/images if the correlation between spectral bands are high
  - Material spectral correlation
  - Topographic shading
  - Sensor bands overlap (adjacent bands)

Spectral Enhancements

Principal Components Analysis (PCA)

Principle Components Analysis

- Adjacent bands in a multispectral remotely-sensed image are generally correlated
- The presence of correlations among the bands of a multispectral image implies that there is redundancy in the data, i.e. some information is being repeated.
- It is the repetition of information between the bands that is reflected in their inter-correlations
- Principal components analysis is a technique that transforms the original remotely sensed dataset into a substantially smaller and easier to interpret set of uncorrelated variables that represents most of the information present in the original dataset

Principle Components Analysis

- Multispectral images may have as few as 3, 4, or 7 bands and other image possibly 200 or more
- With so much data, processing of even modest-sized images requires considerable time
- In this context, band selection assumes considerable practical significance, as image analysts wish to reduce the amount of data while retaining effectiveness
- Variance–covariance and co-relation matrices, shows interrelationships between pairs of bands
- Some pairs show rather strong correlations, and some may not
- Band selection attempts to identify, then remove such duplication so that the dataset can include maximum information using the minimum number of bands

Principle Component Transformation (PCT)

- Method of compressing a lot of data into something that captures the Essence of actual data
- Principle Component Transformation (PCT) is a feature space transformation designed to remove the spectral redundancy
- Principal Component Analysis, de-correlates the data by transforming DN distributions around sets of new multi-spaced axes
- Principal component analysis condenses the information in inter-correlated bands into a few variables, called Principal Components (PCs)
- PCA transformation is scene specific
- Spectral redundancy: A tendency for multiband images if the correlation between spectral bands are high

Principle Component Transformation (PCT)

- **Objective:**
- **To reduce dimensionality of image while maintain information as much possible**
- **Reducing original variables into lower numbers**

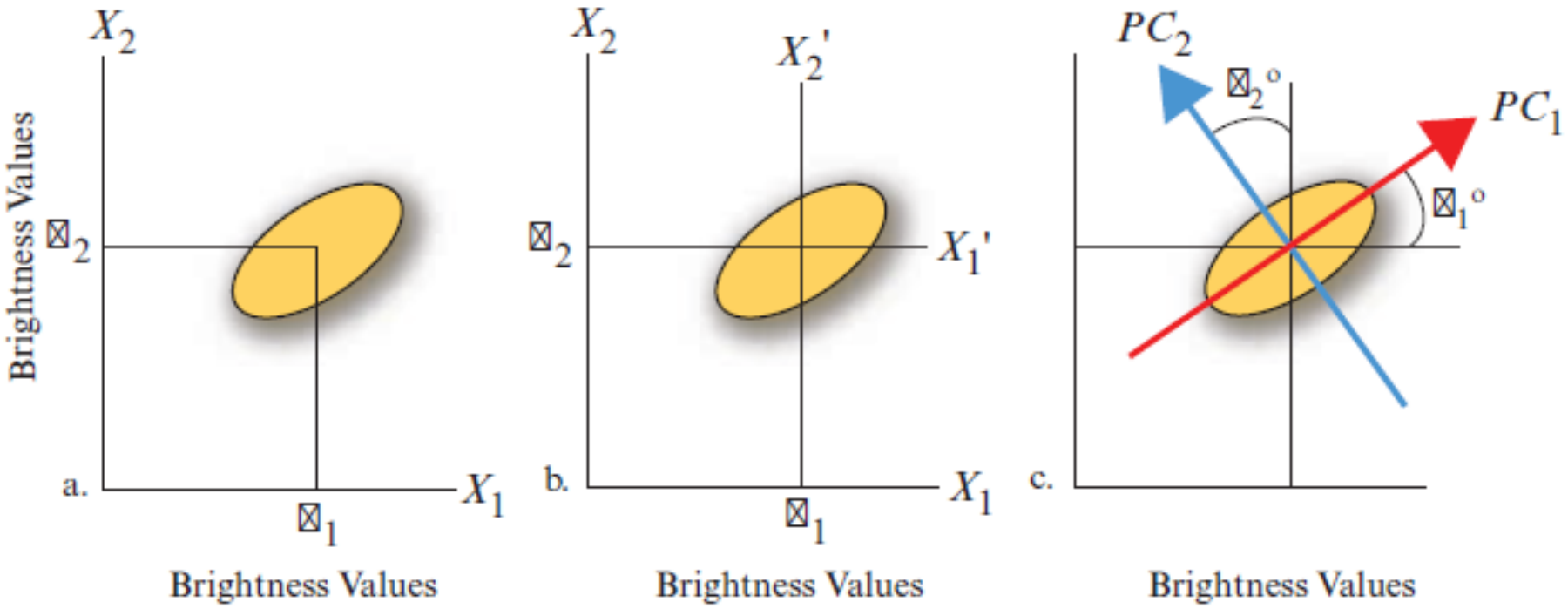
PCA-Purposes

- The analysis of data to identify and find patterns to reduce the dimensions of the dataset with minimal loss of information
- Reduce the computational costs, by extracting a subspace that describes data "best"
- Expressing the data in such a way as to highlight their similarities and differences
- Transformation of the multi band images in new principal component images that may be more interpretable than the original data

Principle Components Analysis

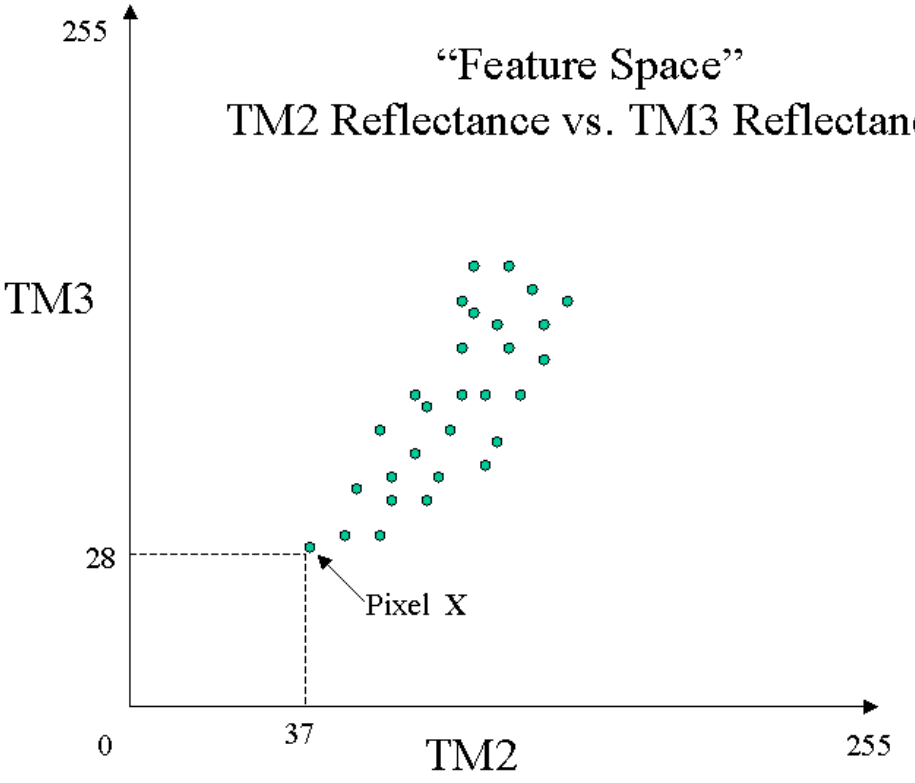
- Principle components analysis rotates the axes of feature space so that they take maximum advantage of data variations
- The first axis (PC1) is put in the direction of maximum variation
- The second axis (PC2) is placed perpendicular to PC1 in the direction of the maximum remaining variation
- Components are ordered in terms of the amount of variance explained, the first two or three components will carry most of the real information of the original data set
- Later components describe only the minor variations

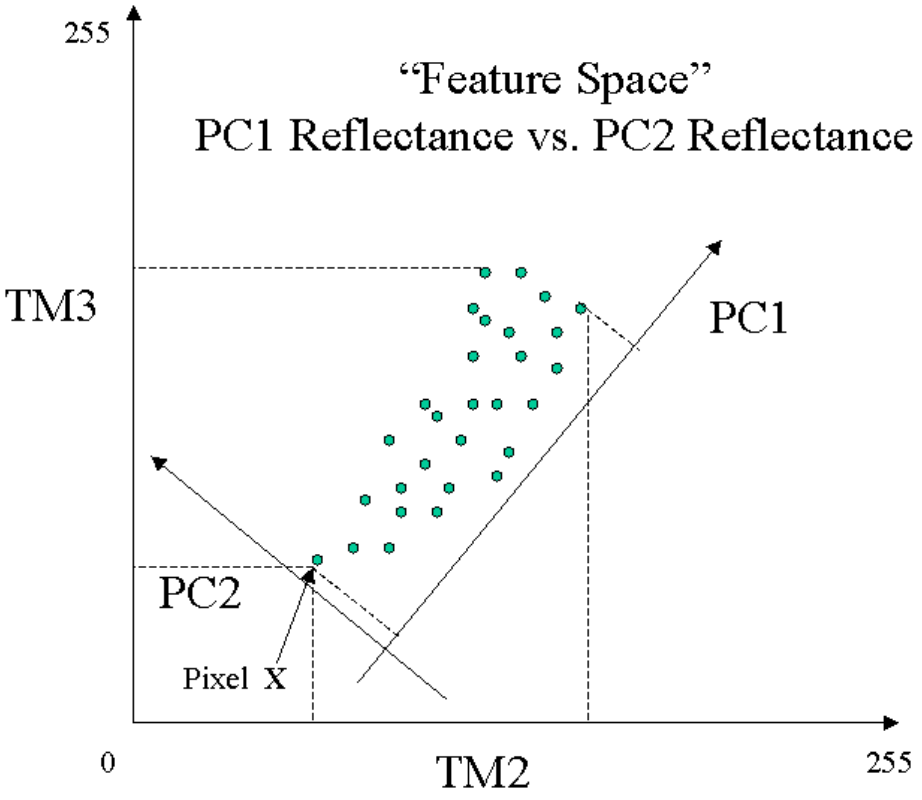
Principal Components Analysis



The spatial relationship between the first two principal components: (a) Scatter-plot of data points collected from two bands labeled X_1 and X_2 with the means of the distribution labeled μ_1 and μ_2. (b) A new coordinate system is created by shifting the axes to an X' system. The values for the new data points are found by the relationship $X_{1'} = X_1 - \mu_1$ and $X_{2'} = X_2 - \mu_2$. (c) The X' axis system is then rotated about its origin (μ_1, μ_2) so that PC_1 is projected through the semi-major axis of the distribution of points and the variance of PC_1 is a maximum. PC_2 must be perpendicular to PC_1. The *PC* axes are the Principal Components of this 2-dimensional data space. Principal Component 1 usually accounts for approximately 90% of the variance, with Principal Component 2 accounting for approximately 5%.

Principle Components Analysis





The way you look at data counts…



The way you look at data counts…



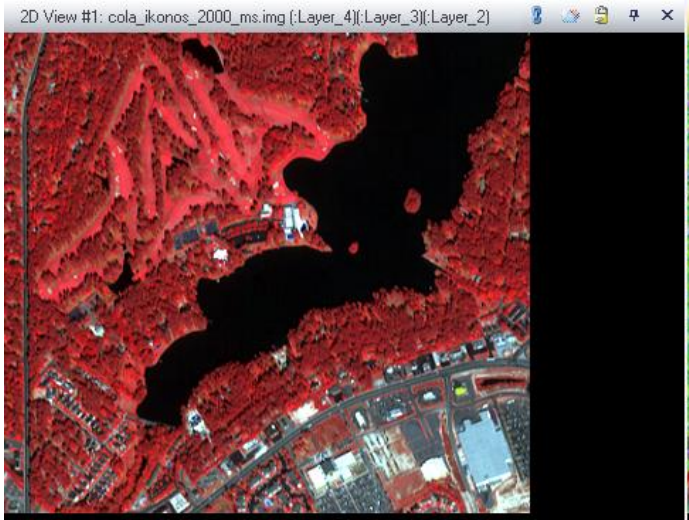
The way you look at data counts…



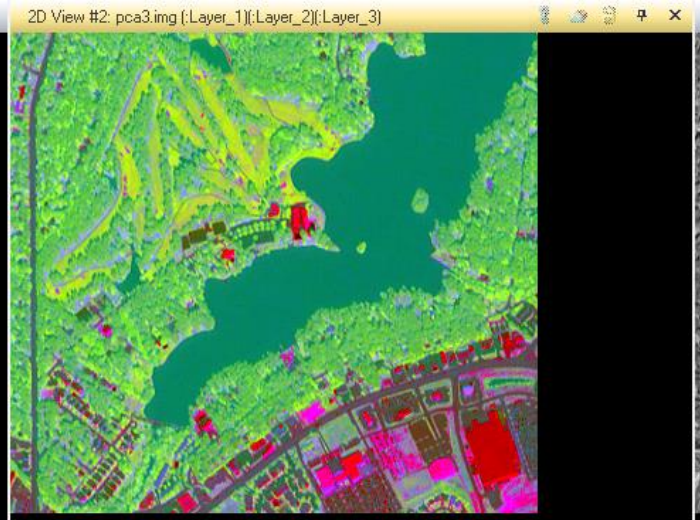
Visual Interpretation of PCA Images

- The information available in the PCA images can be revealed better by combining them visually as registered overlays
- Any three of the PC images can be made into color composites with various assignments of Blue, Green and Red guns on color monitor for better visualization





Original Image



PC Composite Image



PC-1

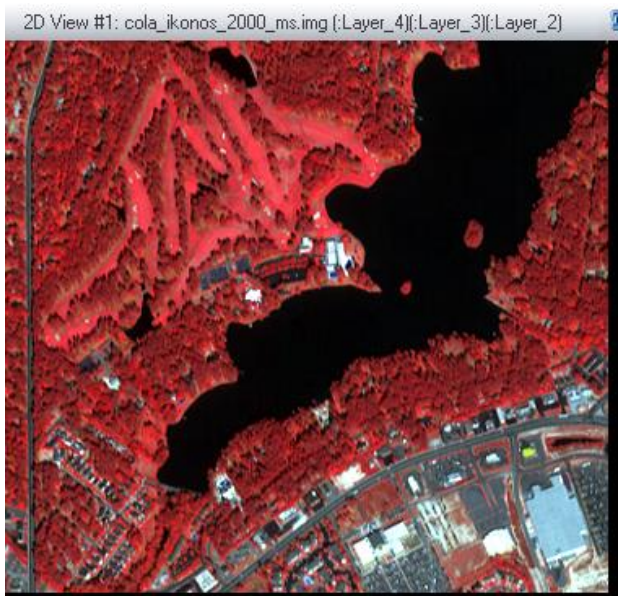
PC-2



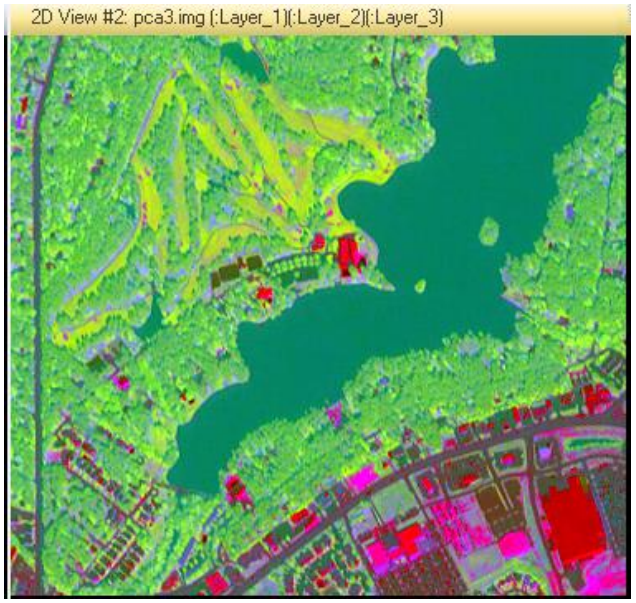
PC-3



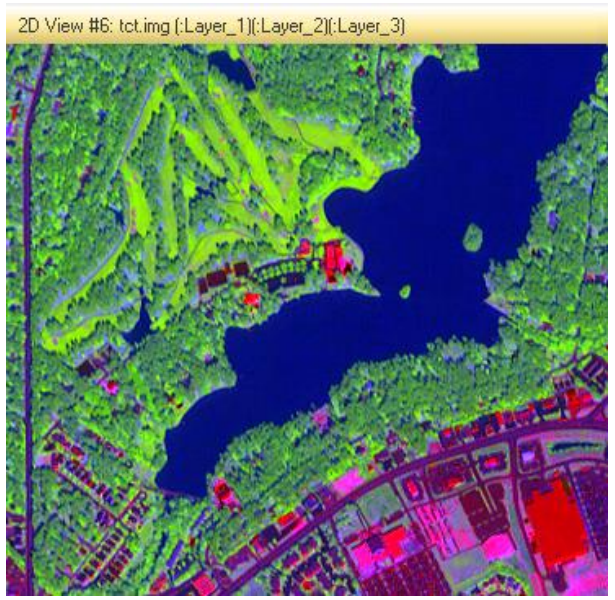
PCA and TCT Component Images



Original Image



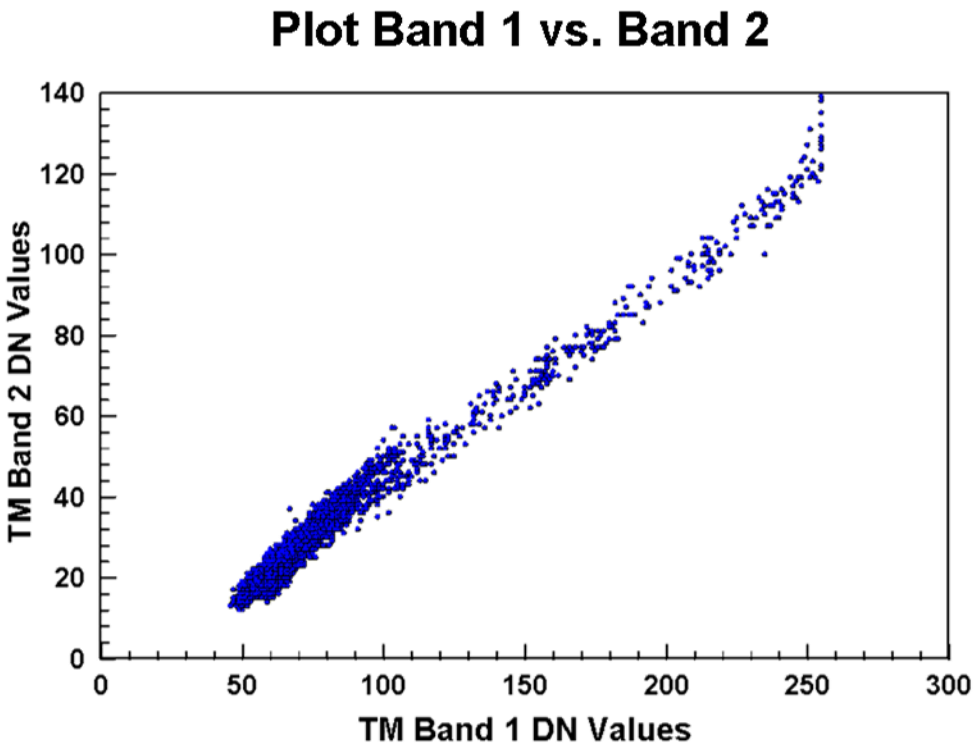
PC Composite Image



TCT Composite Image

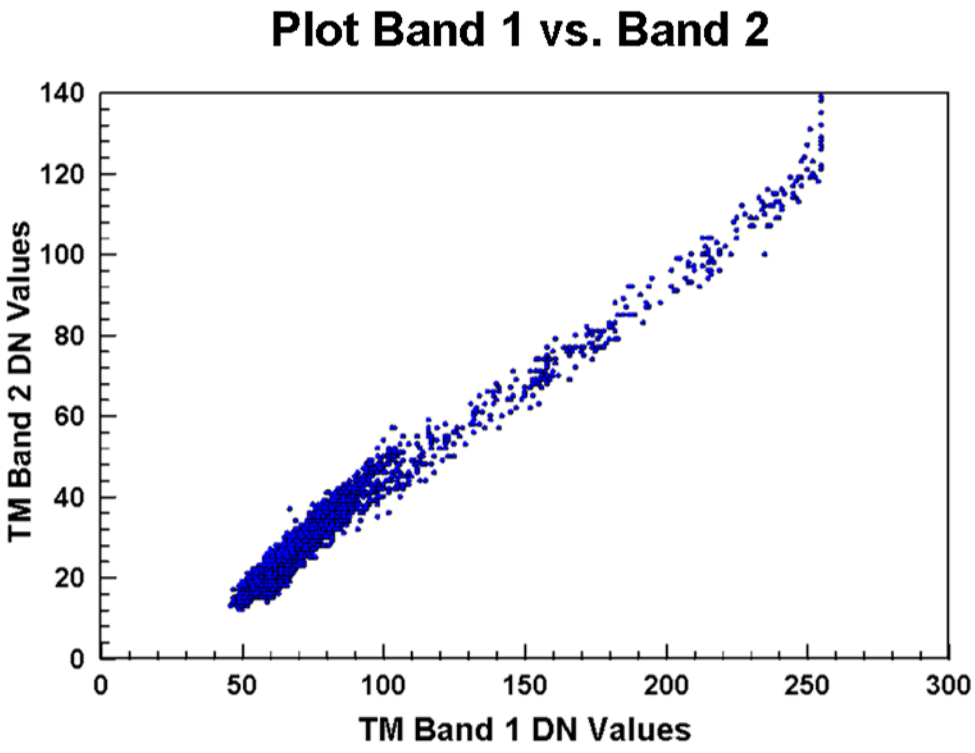
Scatter Plot (Band 1 and Band 2)

- The slightly scattered data form a narrow plot that is almost a straight line
- Can we draw information from his plot???



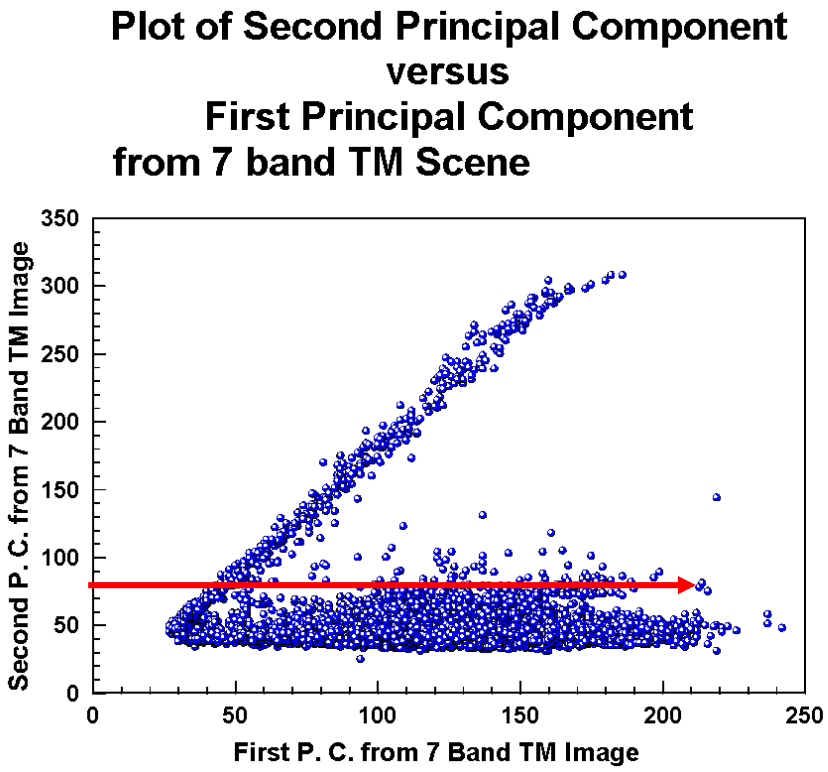
Scatter Plot (Band 1 and Band 2)

- The slightly scattered data form a narrow plot that is almost a straight line
- The two bands are said to be correlated, that is, as Band 1 varies, so does Band 2, and either could be used in place of the other

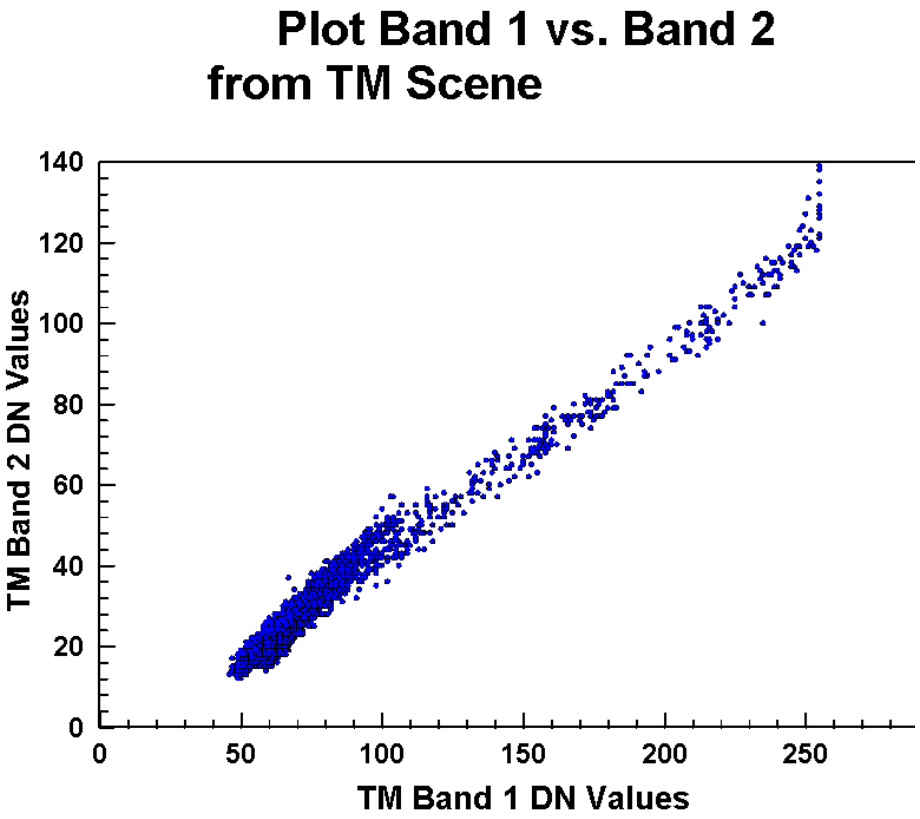


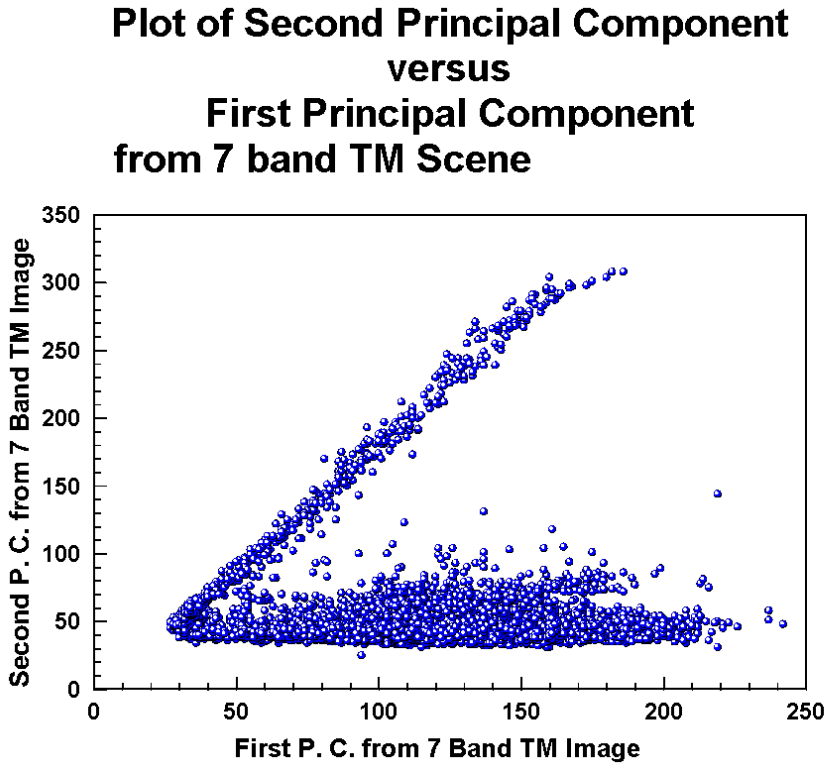
Scatter Plot (PC1 and PC2)

- A plot of the resulting first two components, showing definite dissimilarities
- Note that the distribution of plotted values is spread apart, i.e., the dataset is bimodal



Scatter Plot Original bands VS PCs





Principle Components Analysis

- This concept in two dimensions, but it can be extended to as many dimensions as there are spectral bands

?